9 Rotational Motion

Rotational Dynamic

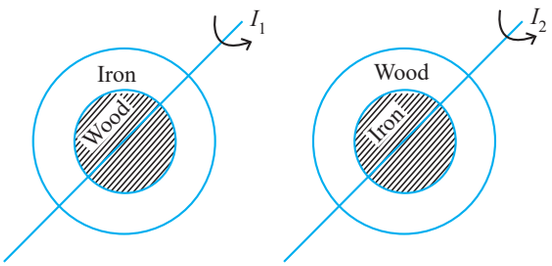
1. A rigid body rotates about a fixed axis with variable angular velocity equal to $\alpha - \beta t$ where t is time and α, β are constants. The angle (in radian) through which it rotates before its stops

(*a*) $\frac{\alpha^2}{2\beta}$ (*b*) $\frac{\alpha^2 - \beta^2}{2\alpha}$

(*c*) $\frac{\alpha^2 - \beta^2}{2\beta}$ (*d*) $\frac{(\alpha - \beta)\alpha}{2}$

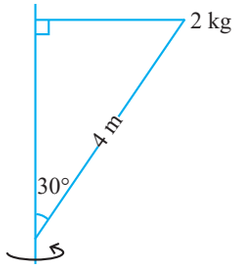
Moment of Inertia

2. Two solid cylindrical objects of equal mass and radius are made from iron and wood, but with different material distributions: The first object has an iron outer body and a wooden core. The second object has a wooden outer body and an iron core. Both rotate about their central axis. Which object has the greater moment of inertia?

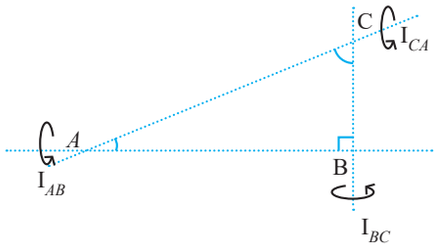


3. A particle of mass 2 kg is attached to a light rod at a distance of 4 m from a fixed axis, as shown in the diagram. The rod makes an angle of 30° with the vertical. Find the perpendicular distance from the axis to the particle.

Using this distance, calculate the moment of inertia of the system about the fixed axis.



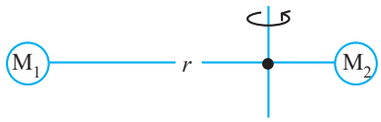
4. Which of the following is correct



(*a*) $I_{AB} = I_{BC} = I_{CA}$ (*b*) $I_{AB} > I_{BC} > I_{CA}$

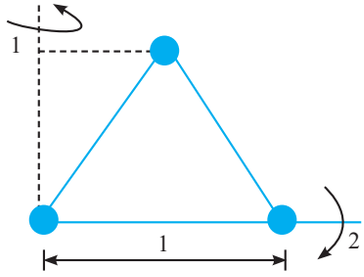
(*c*) $I_{AB} < I_{BC} < I_{CA}$ (*d*) $I_{BC} > I_{AB} > I_{CA}$

5. Two point masses M_1 and M_2 are connected by a massless rod of length r. The system rotates about an axis passing through the center of mass and perpendicular to the rod. Find the moment of inertia (M.O.I) of the system about this axis.



6. Three particles each of mass m are placed at the corners of equilateral triangle of side l

which of the following is/are correct?



(a) Moment of inertia about axis '1' is $\frac{5}{4}ml^2$

(b) Moment of inertia about axis '2' is $\frac{3}{4}ml^2$

(c) Moment of inertia about an axis passing through one corner and perpendicular to the plane is $2\ ml^2$

(d) All of these

7. A thin rod of length L and mass M is bent at its midpoint into two halves so that the angle between them is 90°. The moment of inertia of the bent rod about an axis passing through the bending point and perpendicular to the plane defined by the two halves of the rod is

(a) $\frac{\sqrt{2}ML^2}{24}$ (b) $\frac{ML^2}{24}$ (c) $\frac{ML^2}{12}$ (d) $\frac{ML^2}{6}$

8. Moment of Inertia (M.I.) of four bodies having same mass 'M' and radius '2R' are as follows:

I_1 = M.I. of solid sphere about its diameter

I_2 = M.I. of solid cylinder about its axis

I_3 = M.I. of solid circular disc about its diameter

I_4 = M.I. of thin circular ring about its diameter

If $2(I_2 + I_3) + I_4 = xI_1$, then the value of x will be____.

9. The moment of inertia of a thin uniform circular disc about one of its diameter is I. Its moment of inertia about an axis tangent to it and perpendicular to its plane is

(a) $\frac{2I}{3}$ (b) $2I$ (c) $\frac{I}{2}$ (d) $6I$

10. A thin uniform wire of mass m and length L is bent into a circle. The moment of inertia of the wire about an axis passing through its one end and perpendicular to the plane of the circle is

(a) $\frac{2mL^2}{\pi^2}$ (b) $\frac{mL^2}{\pi^2}$ (c) $\frac{mL^2}{2\pi^2}$ (d) $\frac{mL^2}{3\pi^2}$

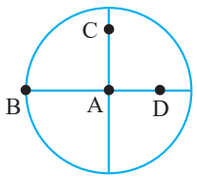
11. A thin wire of length l and mass m is bent in the form of a semicircle as shown. Its moment of inertia about an axis joining its free ends will be

(a) ml^2 (b) zero (c) $\frac{ml^2}{\pi^2}$ (d) $\frac{ml^2}{2\pi^2}$

12. The moment of inertia of a uniform circular disc of radius R and mass M about an axis passing from the edge of the disc and normal to the disc is

(a) $\frac{1}{2}MR^2$ (b) MR^2 (c) $\frac{7}{2}MR^2$ (d) $\frac{3}{2}MR^2$

13. The moment of inertia of a uniform circular disc is maximum about an axis perpendicular to the disc and passing through

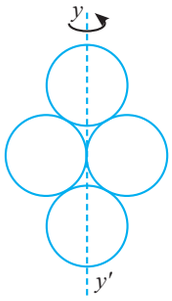


(a) D (b) A (c) B (d) C

14. Two discs of mass and same thickness have densities as 17 g/cm^3 and 51 g/cm^3. The ratio of their moment of inertia about their axes is

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{1}$ (d) $\frac{3}{2}$

15. Four rings each of mass M and radius R are arranged as shown in the figure. The moment of inertia of the system about the axis yy' is



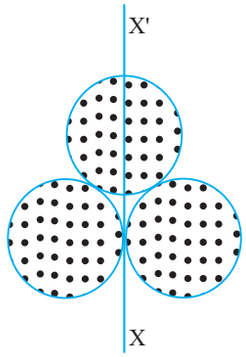
(a) $2\ MR^2$ (b) $3\ MR^2$ (c) $4\ MR^2$ (d) $5\ MR^2$

16. Four thin uniform rods each of length L and mass m are joined to form a square. The moment of inertia of square about an axis along its one diagonal is

(a) $\frac{mL^2}{6}$ (b) $\frac{2}{3}mL^2$ (c) $\frac{3mL^2}{4}$ (d) $\frac{4mL^2}{3}$

17. Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' which is touching to two shells and passing through diameter to third shell.

Moment of inertia of the system consisting of these three spherical shells about XX' axis is



(a) $4\ mr^2$ (b) $\frac{11}{5}mr^2$ (c) $3\ mr^2$ (d) $\frac{16}{5}mr^2$

18. Four spheres of diameter $2a$ and mass M are placed with their centres on the four corners of square of side b. The moment of inertia of the system about an axis about one of the sides of the square is

(a) $Ma^2 + 2Mb^2$ (b) Ma^2

(c) $Ma^2 + 4Mb^2$ (d) $\frac{8}{5}Ma^2 + 2Mb^2$

19. Match column-I with column-II.

Column-I		Column-II	
A.	M.O.I. of soild sphere of radius R about any tangent	I.	$\frac{5}{3}MR^2$
B.	M.O.I. of hollow sphere of radius R about any tangent	II.	$\frac{7}{5}MR^2$
C.	M.O.I. of circular ring of radius R about any diameter	III.	$\frac{MR^2}{4}$
D.	M.O.I. of circular disc of radius R about any diameter	IV.	$\frac{MR^2}{2}$

Choose the correct answer from the options given below:

(*a*) A-III, B-II, C-IV, D-I (*b*) A-III, B-IV, C-II, D-I

(*c*) A-II, B-I, C-IV, D-III (*d*) A-IV, B-II, C-I, D-III

20. A wheel comprises of a ring of radius R and mass M and three spokes of mass m each. The moment of inertia of the wheel about its axis is

(*a*) $\left(M+\frac{m}{4}\right)R^2$ (*b*) $(M+m)R^2$

(*c*) $(M+3m)R^2$ (*d*) $\left(\frac{M+m}{2}\right)R^2$

21. The moment of inertia of a thin uniform rod of mass M and length L about an axis passing through its midpoint and perpendicular to its length is I_0. Its moment of inertia about an axis passing through one of its ends and perpendicular to its length is

(*a*) $I_0 + ML^2/2$ (*b*) $I_0 + ML^2/4$

(*c*) $I_0 + 2\,ML^2$ (*d*) $I_0 + ML^2$

22. Match column-I with column-II.

Column-I		Column-II	
A.	M.O.I of rod (length *l*, mass) about axis ⊥ to rod passing through the midpoint.	(*i*)	$\frac{8ML^2}{3}$
B.	M.O.I. of rod (length 1, mass 2m) about an axis to rod passing through one of its end.	(*ii*)	$\frac{ML^2}{3}$
C.	M.O.I. of rod (length 21, mass m) about an axis to rod passing through its mid-point.	(*iii*)	$\frac{ML^2}{12}$
D.	M.O.I. of rod (length 21, mass 2m) about an axis to rod passing through one of its end.	(*iv*)	$\frac{2ML^2}{3}$

Choose the correct answer from the options given below:

(*a*) A-(*iii*), B-(*ii*), C-(*iv*), D-(*i*)

(*b*) A-(*iii*), B-(*iv*), C-(*ii*), D-(*i*)

(*c*) A-(*ii*), B-(*iii*), C-(*i*), D-(*iv*)

(*d*) A-(*iv*), B-(*ii*), C-(*i*), D-(*iii*)

23. A particle of mass 1 kg is kept at (1 m, 1 m, 1 m). The moment of inertia of this particle about *z*-axis would be

(*a*) 1 kg - m^2 (*b*) 2 kg - m^2

(*c*) 3 kg - m^2 (*d*) None of these

24. From a circular ring of mass *M* and radius *R*, an arc corresponding to a 90° sector is removed The moment of inertia of the remaining part of the ring about an axis passing through the center of the ring and perpendicular to the plane of the ring is *k* times MR^2. Then the value of *k* is

(*a*) $\frac{3}{4}$ (*b*) $\frac{7}{4}$ (*c*) $\frac{3}{2}$ (*d*) $\frac{3}{5}$

25. A T joint is formed by two identical rods A and B each of mass m and length L in the XY plane as shown. Its moment of inertia about axis passing through perpendicular to the plane of the joint.

(*a*) $\frac{2mL^2}{3}$ (*b*) $\frac{mL^2}{12}$

(*c*) $\frac{mL^2}{6}$ (*d*) None of these

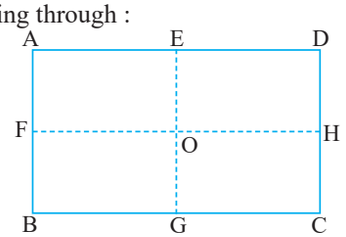
26. Four identical thin rods each of mass M and length *l* form a square frame. Moment of inertia of this frame about an axis through the centre of the square and perpendicular to its plane is

(*a*) $\frac{1}{3}M\ell^2$ (*b*) $\frac{4}{3}M\ell^2$ (*c*) $\frac{2}{3}M\ell^2$ (*d*) $\frac{13}{3}M\ell^2$

27. Three rods each of mass m and length L are joined to form an equilateral triangle as shown in the figure. What is the moment of inertia about an axis passing through the centre of mass of the system and perpendicular to the plane?

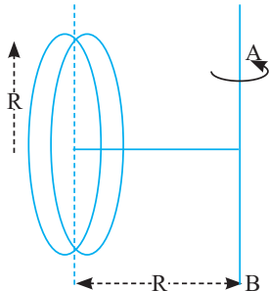
(*a*) $2\,mL^2$ (*b*) $mL^2/2$ (*c*) $mL^2/3$ (*d*) $mL^2/6$

28. In the rectangular lamina shown in the figure, AB = BC/2. The moment of inertia of the lamina is minimum along the axis passing through :



(*a*) AB (*b*) BC (*c*) EG (*d*) FH

29. Find the moment of inertia of the ring shown in figure about the axis AB.



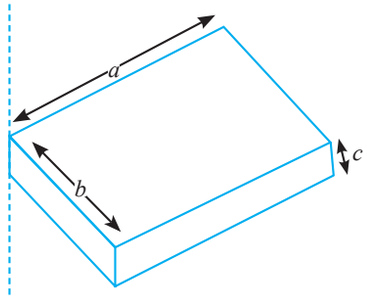
30. The mass density of a rod at a distance x from its end is $\frac{M_0}{2L^2}x$. What is the MOI of the rod of length L about an axis passing through the end of the rod & perpendicular to the rod.

31. Perpendicular axis theorem is valid for

(*a*) disc (*b*) solid sphere

(*c*) Rod (*d*) All of these

32. The figure shows a uniform solid block of mass M and edge lengths *a*, *b* and *c*. Its M.O.I, about an axis through one edge and perpendicular (as shown) to the large face of the block is



(*a*) $\frac{M}{3}(a^2+b^2)$ (*b*) $\frac{M}{4}(a^2+b^2)$

(*c*) $\frac{7M}{12}(a^2+b^2)$ (*d*) $\frac{M}{12}(a^2+b^2)$

33. The moment of inertia of a uniform semicircular wire of mass *m* and radius *r*, about an axis passing through its centre of mass and perpendicular to its plane is

(*a*) $mr^2/2$ (*b*) mr^2

(*c*) $mr^2\left(1-\frac{4}{\pi^2}\right)$ (*d*) $mr^2\left(1+\frac{4}{\pi^2}\right)$

34. The two spheres, one of which is hollow and other solid, have identical masses and moment of inertia about their respective diameters. The ratio of their radii is given by

(*a*) 5 : 7 (*b*) 3 : 5 (*c*) $\sqrt{3}:\sqrt{5}$ (*d*) 3 : 7

35. Three objects, *A* : (a solid sphere), *B*: (a thin circular disk) and *C*: (a circular ring), each have the same mass *M* and radius *R*. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation

(*a*) $W_C > W_B > W_A$ (*b*) $W_A > W_B > W_C$

(*c*) $W_B > W_A > W_C$ (*d*) $W_A > W_C > W_B$

36. The moment of inertia of a uniform cylinder of length *l* and radius *R* about its perpendicular bisector is *I*. What is the ratio *l*/*R* such that the moment of inertia is minimum?

(*a*) 1 (*b*) $\frac{3}{\sqrt{2}}$ (*c*) $\sqrt{\frac{3}{2}}$ (*d*) $\frac{\sqrt{3}}{2}$

37. A square sheet of edge length *L* is used to form a hollow cylinder. The moment of inertia of this cylinder about the central axis is (Where "σ" is surface mass density of sheet);

(*a*) $\frac{2\sigma L^4}{\pi^2}$ (*b*) $\frac{\sigma L^4}{8\pi^2}$ (*c*) $\frac{\sigma L^4}{4\pi^2}$ (*d*) $\frac{\sigma L^4}{3\sqrt{2}\pi^2}$

38. If lamina is in *x*-*y* plane then from theorem of perpendicular axis

A. $I_x - I_y = I_z$ B. $I_x - I_z = I_y$

C. $I_x + I_y = I_z$ D. $I_y + I_z = I_x$

True statements is / are:

(*a*) B and C (*b*) Only D (*c*) Only C (*d*) C and D

39. Four identical rods each of mass *m* and length '*l*' are joined to form a rigid square, centre is at origin and sides are parallel to *x* and y axis then moment of inertia about

i	An axis parallel to z-axis and passing through a corner	P	$\frac{5}{3}ml^2$
ii	One side	Q	$\frac{2}{3}ml^2$
iii	The x-axis	R	$\frac{4}{3}ml^2$
iv	The z-axis passing through centre	S	$\frac{10}{3}ml^2$

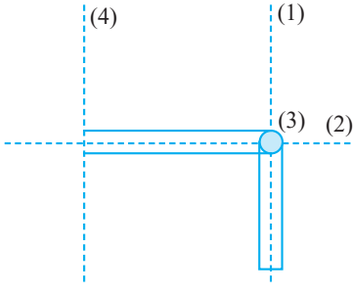
(*a*) i – S, ii – P, iii – Q, iv –

(*b*) i – P, ii – Q, iii – R, iv – S

(*c*) i – Q, ii – R, iii – P, iv –

(*d*) i – S, ii – P, iii – R, iv – Q

40. In diagram axis (3) is perpendicular to plane. Both rods are mass m and length '*l*' then moment of inertia of system



i.	About Axis (1)	P	$\frac{ml^2}{3}$
ii.	About Axis (2)	Q	$\frac{4ml^2}{3}$
iii.	About Axis (3)	R	$\frac{2ml^2}{3}$
iv.	About Axis (4)	S	

(*a*) i – P, ii – P, iii - R, iv – Q

(*b*) i – Q, ii – Q, iii - P, iv – R

(*c*) i – R, ii – R, iii - P, iv – Q

(*d*) i – P, ii – R, iii - Q, iv – P

41. Three objects [A solid sphere a thin circular disc and circular ring] each have same mass M but have different radius R_S, R_D and R_R respectively. They all spin with same angular speed ω about their own symmetric axes. If the amount of work (W) required to bring them to rest are same. Find x if

$$\frac{R_S^2 + R_D^2}{R_R^2} = \frac{x}{2}.$$

42. From a circular disc of radius R and mass 9M, a small disc of mass M and radius R/3 is removed concentrically. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through its centre is

(*a*) $\frac{40}{9}MR^2$ (*b*) MR^2 (*c*) $4MR^2$ (*d*) $\frac{4}{9}MR^2$

43. From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through centre?

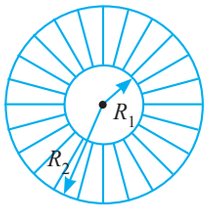
(*a*) 15 $MR^2/32$ (*b*) 13 $MR^2/32$

(*c*) 11 $MR^2/32$ (*d*) 9 $MR^2/32$

44. From a circular disc of radius R and mass 9M, a small disc of mass M and radius $R/3$ is removed concentrically. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through its centre is

(*a*) $\frac{40}{9}MR^2$ (*b*) MR^2 (*c*) $4MR^2$ (*d*) $\frac{4}{9}MR^2$

45. A disc of radius R_1 is cut out from a disc of radius R_2 as shown in figure. If the disc left has mass M then find the moment of inertia of the disc left about an axis passing through its centre and perpendicular to its plane.



(*a*) $\frac{M}{2}(R_2^3 - R_1^2)$ (*b*) $\frac{M}{2}(R_2^2 + R_1^2)$

(*c*) $\frac{M}{2}\left(\frac{R_2 + R_1}{2}\right)^2$ (*d*) $\frac{MR_2^2}{2}$

46. **Assertion (A):** An electric fan continue to rotate for some time after the current is switched off.

Reason (R): Fan continues to rotate due to inertia of motion.

(*a*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

(*b*) Assertion (A) is true but Reason (R) is false

(*c*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(*d*) Assertion (A) is false but Reason (R) is true

Radius of Gyration

47. Two rings of same mass and radius R are placed with their planes perpendicular to each other and centres at a common point. The radius of gyration of the system about an axis passing through the centre and perpendicular to the plane of one ring is

(*a*) $2R$ (*b*) $R/\sqrt{2}$ (*c*) $\sqrt{\frac{3}{2}}R$ (*d*) $\frac{\sqrt{3}R}{2}$

48. Four particles each of mass m are placed at the corners of a square of side length t The radius of gyration of the system about an axis perpendicular to the square and passing through centre is:

(*a*) $\frac{\ell}{\sqrt{2}}$ (*b*) $\frac{\ell}{2}$ (*c*) ℓ (*d*) $\ell\sqrt{2}$

49. The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is

(*a*) 2 : 1 (*b*) $\sqrt{5}:\sqrt{6}$ (*c*) 2 : 3 (*d*) $1:\sqrt{2}$

50. The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the centre will be ____m. Given, the length of the rod is $10\sqrt{3}\,m$.

51. (1) Centre of gravity (C.G.) of a body is the point at which the weight of the body acts.

(2) Centre of mass coincides with the centre of gravity if the earth is assumed to have infinitely large radius.

(3) To evaluate the gravitational field intensity due to any body at an external point, the entire mass of the body can be considered to be concentrated at its C.G.

(4) The radius of gyration of any body rotating about an axis is the length of the perpendicular drawn from the C.G. of the body to the axis.

Which one of the following pairs of statements is correct?

(*a*) (4) and (1) (*b*) (1) and (2)

(*c*) (2) and (3) (*d*) (3) and (4)

52. **Assertion (A):** MOI of circular diss of mass 'M' and radius 'R' about x, y-axis (passing through its plane) and z-axis which is perpendicular to its plane work found to be I_x, I_y and I, respectively. The radii of gyration about all three axis will be same.

Reason (R): A rigid body making rotational motion has fixed mass and shape.

(*a*) Both Assertion (A) and Reason (R) arc true but Reason (R) is not the correct explanation of Assertion (A)

(*b*) Assertion (A) is true but Reason (R) is false

(*c*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(*d*) Assertion (A) is false but Reason (R) is true

Torque (Moment of Force)

53. Find the torque about the origin when a force of $3\hat{j}\,N$ acts on a particle whose position vector is $2\hat{k}\,m$

(*a*) $6\hat{i}\,N\,m$ (*b*) $6\hat{j}\,N\,m$

(*c*) $-6\hat{i}\,N\,m$ (*d*) $6\hat{k}\,N\,m$

54. The torque about the origin when a force of $3\hat{j}\,N$ acts on a particle is $-6\hat{i}$ Nm. Find position vector of particle.

(*a*) $-2\hat{k}\,m$ (*b*) $2\hat{k}\,m$ (*c*) $18\hat{k}\,m$ (*d*) $-18\hat{k}\,m$

55. The moment of the force, $\vec{F}=4\hat{i}+5\hat{j}-6\hat{k}$ at (2, 0, –3), about the point (2, –2, –2) is given by

(*a*) $-8\hat{i}-4\hat{j}-7\hat{k}$ (*b*) $-4\hat{i}-\hat{j}-8\hat{k}$

(*c*) $-7\hat{i}-8\hat{j}-4\hat{k}$ (*d*) $-7\hat{i}-4\hat{j}-8\hat{k}$

56. Moment of a force of magnitude 20 N acting along positive *x*-direction at point (3m, 0, 0) about the point (0, 2, 0) (in N-m) is

(*a*) 20 (*b*) 60 (*c*) 40 (*d*) 30

57. What is the torque of the force vec $\vec{F}=2\hat{i}-3\hat{j}+4\hat{k}\,N$ acting at the point vec $\vec{r}=3\hat{i}+2\hat{j}+3\hat{k}\,m$ km about origin?

(*a*) $-6\hat{i}+6\hat{j}-12\hat{k}$ (*b*) $-17\hat{i}+6\hat{j}-13\hat{k}$

(*c*) $6\hat{i}-6\hat{j}+12\hat{k}$ (*d*) $17\hat{i}-6\hat{j}-13\hat{k}$

58. If $\vec{F}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then

(*a*) $\vec{r}\cdot\vec{\tau}>0$ and $\vec{F}\cdot\vec{r}<0$ (*b*) $\vec{r}\cdot\vec{\tau}=0$ and $\vec{F}\cdot\vec{r}=0$

(*c*) $\vec{r}\cdot\vec{\tau}=0$ and $\vec{F}\cdot\vec{r}\neq 0$ (*d*) $\vec{r}\cdot\vec{\tau}\neq 0$ and $\vec{F}\cdot\vec{r}=0$

59. (*i*) A particle of mass m has been thrown up with initial speed *u* making angle θ with the horizontal. Find the torque of its weight about the point of projection when it just reached the highest point.

(*ii*) Determine the torque due to gravitational force about the origin when the object is just about to collide with the ground (i.e., when it reaches the lowest point of its trajectory).

(*iii*) Find the torque due to gravitational force about the origin at any time *t* during the Object's motion.

60. ABC is an equilateral triangle with O as its centre $\vec{F}_1,\vec{F}_2$ *and* $\vec{F}_3$ represent three forces acting along the sides *AB*, *BC* and *AC* respectively. If the total torque about O is zero then the magnitude of $\vec{F}_3$ is

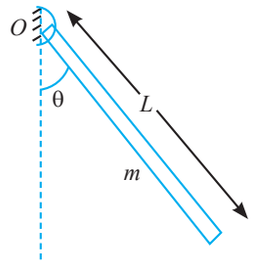
(*a*) $\dfrac{F_1+F_2}{2}$ (*b*) $2(F_1+F_2)$

(*c*) F_1+F_2 (*d*) F_1-F_2

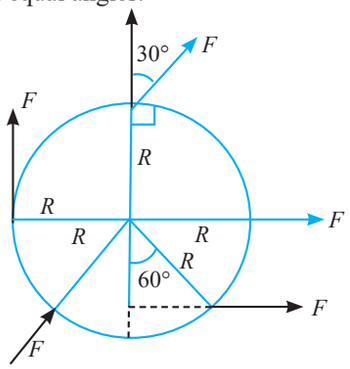
61. A statement is given: "If a body is not rotating, then the torque about every point will be zero." Is this statement true or false?

(*a*) True (*b*) False

62. A uniform rod of length L and mass m is placed at an angle θ from the vertical. The rod is pivoted at point O, then calculate the torque due to the gravitational force about point O when the rod is at this angle θ.



63. Given a disc with radius *R* and multiple forces of equal magnitude *F* acting at different points on the disc, as shown in the diagram, determine the total torque about the center *O*. Assume all forces lie in the plane of the disc and are applied at equal angles.



64. A solid cylinder of mass 50 kg and radius 0.5 m is free to rotate about the horizontal axis. A massless string is wound round the cylinder with one end attached to it and other hanging freely. Tension in the string required to produce an angular acceleration of 2 revolutions s^{-2} is

(*a*) 25 N (*b*) 50 N (*c*) 78.5 N (*d*) 157 M

65. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop it after 2π revolutions is

(*a*) 2×10^{6} N m (*b*) 2×10^{-6} N m

(*c*) 2×10^{-3} N m (*d*) 12×10^{-4} N m

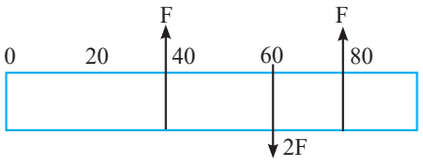
66. A flywheel of mass 6000 kg starts from rest and gets up to a speed of 20 rad/s in 2 minutes. The radius of gyration of the wheel is 100 cm. The average torque exerted on the fly wheel is

(*a*) 1000 Nm (*b*) 750 Nm (*c*) 500 Nm (*d*) 2000 Nm

67. The grinding stone of a flour mill is rotating at 600 rad/sec, for this power is 1.2 k watt Is used. The effective torque on stone in N-m will be

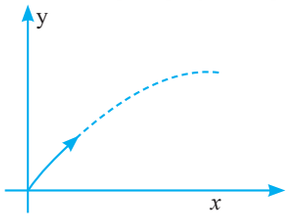
(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4

68. Three parallel forces are acting on a rod at distances of 40 cm, 60 cm and 80 cm respectively from one end of the rod (as shown in figure). Assuming no other forces, the rod under the influence of these forces will:



(*a*) be at rest
(*b*) exhibit rotational motion only
(*c*) exhibit translational motion
(*d*) exhibit combined rotational and translational motion

69. A particle is projected from point 'O' having mass 'm' as shown in figure, moving on projectile path

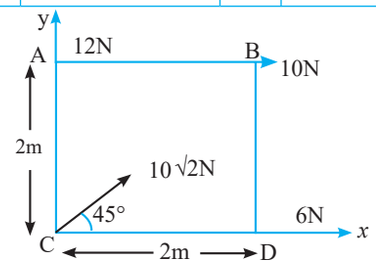


i	Torque about 'O'	P	Zero
ii	Torque about x - axis	Q	Constant non zero
iii	Angular momentum about O	R	Increasing
iv	Angular momentum about x-axis	S	Decreasing

(*a*) i – P, ii – Q, iii – R, iv – S
(*b*) i – R, ii – P, iii – R, iv – P
(*c*) i – Q, ii – R, iii – R, iv – P
(*d*) i – P, ii – R, iii – R, iv – S

70. Value of net torque about given points.

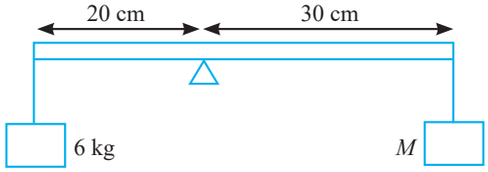
i	Torque about A	P	$-64\hat{k}$
ii	Torque about B	Q	$32\hat{k}$
iii	Torque about C	R	$-12\hat{k}$
iv	Torque about D	S	$-20\hat{k}$



(*a*) i – P, ii – Q, iii – R, iv – S
(*b*) i – Q, ii – P, iii – P, iv – S
(*c*) i – Q, ii – R, iii – S, iv – S
(*d*) i – R, ii – S, iii – P, iv – S

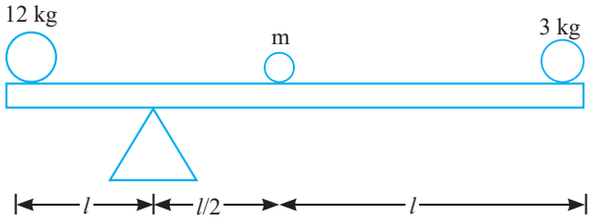
Translational & Rotational Equilibrium

71. The value of M, as shown, for which the rod will be in equilibrium is (rod is massless)



(*a*) 1 kg (*b*) 2 kg (*c*) 4 kg (*d*) 6 kg

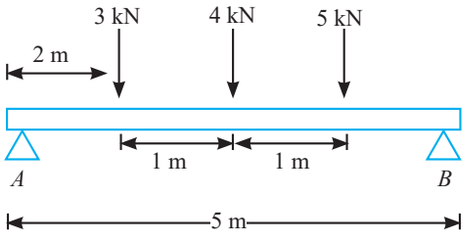
72. For equilibrium of the system, value of mass m should be



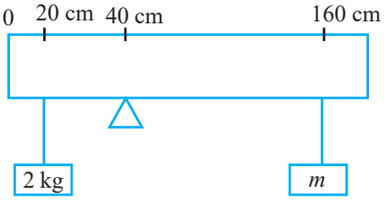
(*a*) 9 kg (*b*) 15 kg (*c*) 21 kg (*d*) 1 kg

73. A Couple produces
(*a*) Linear and rotational motion
(*b*) No motion
(*c*) Purely linear motion
(*d*) Purely rotational motion

74. A massless beam of length 5 m is placed on two wedges *A* and *B*. If three forces 3 kN, 4 kN and 5 kN are applied on the beam as shown in the figure, then find normal reaction at *A* and *B*.



75. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium, ($g = 10\ m/s^2$)



76. Two like parallel forces 20 N and 30 N act at the ends A and B of a rod 1.5 m long. The resultant of the forces will act at the point

(*a*) 90 cm from A (*b*) 75 cm from B
(*c*) 20 cm from B (*d*) 85 cm from A

77. A rod of weight W is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at distance x from A. The normal reaction on A is

(*a*) $\frac{W(d-x)}{d}$ (*b*) $\frac{Wx}{d}$ (*c*) $\frac{Wd}{x}$ (*d*) $\frac{W(d-x)}{x}$

78. What is the angular acceleration of a particle if the angular velocity of *a* particle becomes 4 times of its initial angular velocity 1 rad/s in 2 seconds?

79. A particle moving with an angular velocity of 200 rad s^{-1} starts decelerating at a rate of 2 rad s^{-2}. Calculate the time in which it will come to rest.

80. A fan is rotating with a speed of 450 rev/minute. After being switched off it comes to rest in 10s. Assuming constant angular deceleration, calculate the number of revolutions made by it before coming to rest.

81. A disc rotating about its axis from rest, acquires the angular speed 100* reve/s in 4 second. The angle rotated by it during these four seconds (in radian) is

(*a*) 100 π (*b*) 200 π (*c*) 300 π (*d*) 400 π

82. A wheel has angular acceleration of 3 rad/s^2 and an initial angular speed of 2 rad/s. In a time of two second it has rotated through an angle (in radians) of

(*a*) 10 (*b*) 12 (*c*) 4 (*d*) 6

83. The instantaneous angular position of a point on a rotating wheel is given by the equation $\theta(t) = 2t^3 - 6t^2$. The torque on the wheel becomes zero at:-

(*a*) $t = 1$s (*b*) $t = 0.5$ s (*c*) $t = 0.25$ s (*d*) $t = 2$s

84. A body rotating with uniform angular acceleration covers 100π (radian) in the first 5s after the start. Its angular speed at the end of 5s (in radian/s) is

(*a*) 40 π (*b*) 30 π (*c*) 20 π (*d*) 10 π

85. A constant torque of 100 Nm turns a disc of mass 3 kg and radius 20 m about an axis passing through it's diameter. Starting from rest, its angular velocity after 3s is :

86. A wheel starting from rest is uniformly accelerated at 2 rad/s^2 for 20 seconds. It is allowed to rotate uniformly for the next 10 seconds and is finally brought to rest in next 20 seconds. The total angle rotated by the wheel (in radian) is

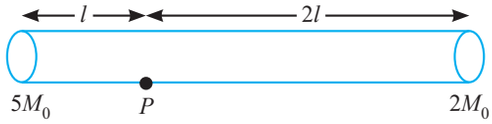
(*a*) 600 (*b*) 1200 (*c*) 300 (*d*) 1800

87. A body rotates about a fixed axis with an angular acceleration of 3 rad/s^2. The angle rotated by it during the time when its angular velocity increases from 10 rad/s to 20 rad/s (in radian) is

(*a*) 50 (*b*) 100 (*c*) 150 (*d*) 200

88. Two particles start moving from same position on a circle of radius 20 cm with speed 40π m/s and 36π m/s respectively in the same direction. Find the time after which the particles will meet again.

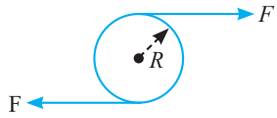
89. A rigid massless road of length $3l$ has tow masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be



(*a*) $\frac{g}{13l}$ (*b*) $\frac{g}{3l}$ (*c*) $\frac{g}{2l}$ (*d*) $\frac{7g}{3l}$

90. The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is_____ (Assuming the acceleration to be uniform).

91. Two equal and opposite forces are applied tangentially to a uniform disc of mass M and radius R as shown in the figure. If the disc is pivoted at its centre and free to rotate in its plane, the angular acceleration of the disc is



(*a*) $\frac{F}{MR}$ (*b*) $\frac{2F}{3MR}$ (*c*) $\frac{4F}{MR}$ (*d*) Zero

92. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about the axis. The torque which can stop the wheel's rotation in one minute would be:

(*a*) $\frac{\pi}{12}N-m$ (*b*) $\frac{\pi}{15}N-m$

(*c*) $\frac{\pi}{18}N-m$ (*d*) $\frac{2\pi}{15}N-m$

93. A wheel has moment of inertia 5×10^{-3} kg m^2 and is making 20 rev/sec. The torque needed to stop it in 10 sec is.....× 10^{-2} N-m :

(*a*) 2π (*b*) 2.5π (*c*) 4π (*d*) 4.5π

94. A solid cylinder of mass 50 kg and radius 0.5 m is free to rotate about the horizontal axis. A massless string is wound round the cylinder with one end attached to it and other hanging freely. Tension in the string required to produce an angular acceleration of 2 revolutions S^{-2} is :

(*a*) 25 N (*b*) 50 N (*c*) 78.5 N (*d*) 157 N

95. A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

(*a*) 25 m/s^2 (*b*) 0.25 rad/s^2

(*c*) 25 rad/s^2 (*d*) 5 m/s^2

96. A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad s^{-2}. Its net acceleration in ms^{-2} at the end of 2.0 s is approximately:

(*a*) 8.0 (*b*) 7.0 (*c*) 6.0 (*d*) 3.0

97. A wheel has angular acceleration of 3 rad /sec^2 and an initial angular speed of 2 rad/sec. In a time of 2 sec it has rotated through an angle (in radian) of

(*a*) 4 (*b*) 6

(*c*) 10 (*d*) 12

98. A solid body rotates about a fixed axis such that its angular velocity depends on e as $\omega = k\theta^{-1}$ where k is a positive constants. At t = 0, $\theta = 0$, then time dependence of θ is given as

(*a*) $\theta = kt$ (*b*) $\theta = 2kt$

(*c*) $\theta = \sqrt{kt}$ (*d*) $\theta = \sqrt{2kt}$

99. A wheel having moment of inertia 4 kg m^2 about its symmetrical axis, rotates at rate of 240 rpm about it. The torque which can stop the rotation of the wheel in one minute is

(*a*) $\frac{5\pi}{7} Nm$ (*b*) $\frac{8\pi}{15} Nm$

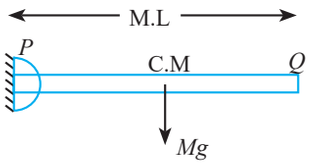
(*c*) $\frac{2\pi}{9} Nm$ (*d*) $\frac{3\pi}{7} Nm$

100. A rod *PQ* of mass *M* and length *L* is hinged at end *P*. The rod is kept horizontal by a massless string tied to point *Q* as shown in figure. When string is cut, the initial angular acceleration of the rod is

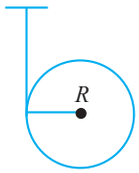
(*a*) $\frac{2g}{L}$ (*b*) $\frac{2g}{2L}$

(*c*) $\frac{3g}{2L}$ (*d*) $\frac{g}{L}$

101. Find accelerating of C.O.M, point *P* and Point *Q* when rod is just released, also find normal reaction at that point *P*.



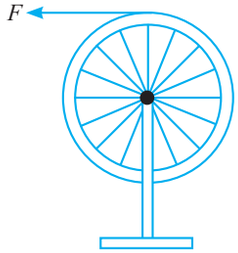
102. A solid cylinder of M has a string wrapped several times around its circumference. The free end of string is attached to the ceiling and the cylinder is released from rest. Find the acceleration of the cylinder and the tension in the string.



(*a*) $a = \frac{2}{3}g$ and $T = \frac{Mg}{3}$ (*b*) $a = \frac{2}{3}g$ and $T = \frac{2Mg}{3}$

(*c*) $a = \frac{5}{3}g$ and $T = \frac{2Mg}{3}$ (*d*) $a = \frac{5}{3}g$ and $T = \frac{Mg}{3}$

103. A string is wrapped around the rim of a wheel of moment of inertia 0.20 kg-m^2 and radius 20 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of F = 20 N. Find the angular velocity of the wheel after 5 s.

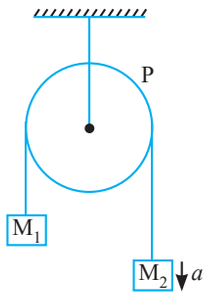


104. A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

(*a*) 0.25 rad s^{-2} (*b*) 25 rad s^{-2}

(*c*) 5 m s^{-2} (*d*) 25 m s^{-2}

105. In the figure, the blocks have unequal masses m_1 and m_2 ($m_1 < m_2$). M_2 has a downward acceleration *a*. The pulley P has a radius r, and some mass.The string does not slip on the pulley. Find the acceleration of block and angular acceleration of pulley. (I = momentum of inertia of pulley)



Mechanical Advantage

106. Which of the following statements are correct?

(1) Centre of mass of a body always coincides with the centre of gravity of the body.

(2) Centre of mass of a body is the point at which the total gravitational torque on the body is zero.

(3) A couple on a body produce both translational and rotational motion in a body.

(4) Mechanical advantage greater than one means that small effort can be used to lift a large load.

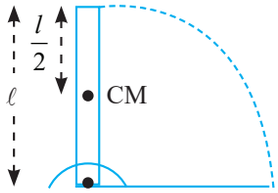
(*a*) (2) and (4) (*b*) (1) and (2)

(*c*) (2) and (3) (*d*) (3) and (4)

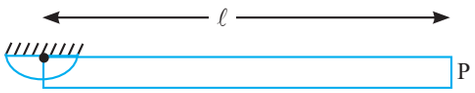
107. If the parallel forces acting on a lever are in the ratio 3 : 5 then what is the mechanical advantage of the lever?

C.O.M.E in Pure Rotational

108. A thin meter scale is kept vertical by placing its lower end hinged on floor. It is allowed to fall. Calculate the velocity of its upper end when it hits the floor.



109. A uniform rod smoothly pivoted at one of its ends is released from rest. If it swings in vertical plane, the maximum speed of the end P of the rod is:



(*a*) $2\sqrt{3g\ell}$ (*b*) $\sqrt{3g\ell}$ (*c*) $2\sqrt{2g\ell}$ (*d*) $\sqrt{g\ell}$

Work

110. An energy of 484 joule is spent in increasing the speed of flywheel by 6 time it's initial value of 60 rpm. The moment of inertia of flywheel is:

(*a*) 0.07 kg m^2 (*b*) 0.7 kg m^2
(*c*) 3.22 kg m^2 (*d*) 30.8 kg m^2

Translational & Rotation

111. Only if magnitude of Position vector is changing then body is in Pure translational motion

(*a*) True (*b*) False

112. Only if direction of position vector is changing then body is in pure rotational motion

(*a*) True (*b*) False

Pure Rolling

113. When a body is under pure rolling, the fraction of its total kinetic energy $\frac{2}{5}$. which is the purely rotational is Identify the body.

114. The kinetic energy of a rolling solid cylinder of mass 4 kg about its axis of rotation is 20 kg m^2 s^{-2}. Calculate the velocity with which it is moving.

115. Which of the following (if mass and radius are assumed to be same) have maximum percentage of total K.E. in rotational form while pure rolling?

(*a*) Disc (*b*) Sphere
(*c*) Ring (*d*) Hollow sphere

116. A spherical shell rolls on a table without slipping. What is the percentage of its K.E. which is rotational?

(*a*) 50% (*b*) 40% (*c*) 70% (*d*) 90%

117. When a sphere of moment of inertia I rolls down on an inclined plane the percentage of total energy which is rotational, is approximately

(*a*) 28% (*b*) 72%
(*c*) 100% (*d*) None of these

118. A discs rolling on an inclined plane without slipping then what fraction of its total energy will be in form of rotational kinetic energy:

(*a*) 1 : 3 (*b*) 1 : 2 (*c*) 2 : 7 (*d*) 2 : 5

119. If rotational kinetic energy is 50% of total kinetic energy then the body will be:

(*a*) Ring (*b*) Cylinder
(*c*) Hollow sphere (*d*) Solid sphere

120. A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it?

(*a*) 3J (*b*) 30kJ (*c*) 2J (*d*) 1J

121. A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio K_t / ($K_t + K_r$) for the sphere is?

(*a*) 10 : 7 (*b*) 7 : 10 (*c*) 2 : 5 (*d*) 5 : 7

122. A circular disc of mass 2 kg and radius 10 cm rolls without slipping with a speed 2 m/s. The total kinetic energy of disc is.

123. A solid cylinder of mass 20 kg rotates about its axis with angular speed 100 rad s^{-1}. The radius of the cylinder is 0.25 m. What is the kinetic energy associated with the rotation of the cylinder?

124. A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K. If radius of the ball be R, then the fraction of total energy associated with its rotational energy will be

(*a*) $\frac{K^2+R^2}{R^2}$ (*b*) $\frac{K^2}{R^2}$

(*c*) $\frac{K^2}{K^2+R^2}$ (*d*) $\frac{R^2}{K^2+R^2}$

125. A solid spherical ball rolls on a table. Ratio of its rotational kinetic energy to total kinetic energy is

(*a*) $\frac{1}{2}$ (*b*) $\frac{1}{6}$ (*c*) $\frac{7}{10}$ (*d*) $\frac{2}{7}$

126. Which of the following (if mass and radius are assumed to be same) have maximum percentage of total K.E. in rotational form while pure rolling?

(*a*) Disc (*b*) Sphere
(*c*) Ring (*d*) Hollow sphere

127. A solid spherical ball is rolling without slipping down an inclined plane. The fraction of its total energy associated with rotation is

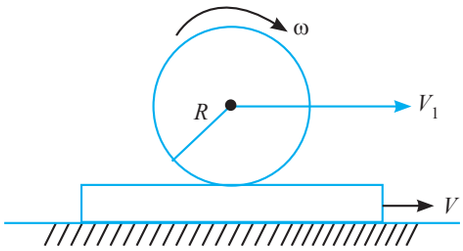
(*a*) $\frac{2}{5}$ (*b*) $\frac{2}{7}$ (*c*) $\frac{3}{5}$ (*d*) $\frac{3}{7}$

128. An object is rolling without slipping on a horizontal surface and its rotational kinetic energy is two-third of translational kinetic energy. The body may be.

(*a*) Disc (*b*) Solid sphere

(*c*) Hollow cylinder (*d*) Spherical shell

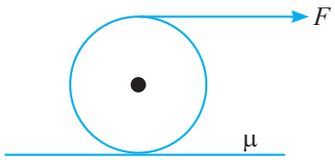
129. A body is rolling without slipping on a moving plank as shown in figure. Relation of v and v_1 is



(*a*) $v_1 - v = R\omega$ (*b*) $v_1 + v = R\omega$

(*c*) $v_1 + v = -R\omega$ (*d*) None of these

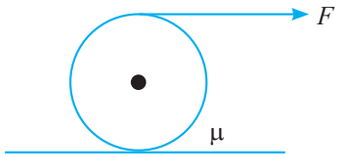
130. A force F is applied on a Hollow cylinder as shown in the figure. For pure accelerated rolling, the force of friction will be towards



(*a*) Forward direction (*b*) Backward direction

(*c*) Normal to the surface (*d*) Zero

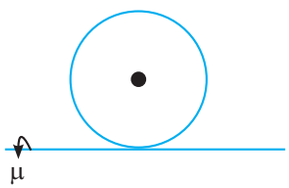
131. A force F is applied on a DISC as shown in the figure. For pure accelerated rolling, the force of friction will be towards



(*a*) Forward direction (*b*) Backward direction

(*c*) Normal to the surface (*d*) Zero

132. A hollow sphere placed on rough surface then find acceleration and value of friction.



133. Ratio of translational kinetic energy to the total kinetic energy of Rolling solid sphere is $x/7$ and ratio of translational kinetic energy to the rotational kinetic energy for a rolling disc is y/z. Find $x + y$.

134. A solid sphere is released from rest on the incline having friction coefficient $\frac{1}{7}\tan\theta$

(*a*) it will start pure rolling,

(*b*) it will make pure translation motion

(*c*) it will start rolling with slipping

(*d*) the angular momentum of the sphere about its centre will remain constant

Rolling Motion on Smooth Inclined Plane

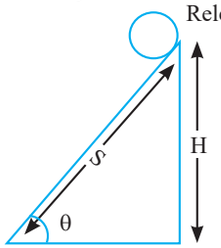
135. On smooth Inclined, pure rolling motion is not possible

(*a*) True (*b*) False

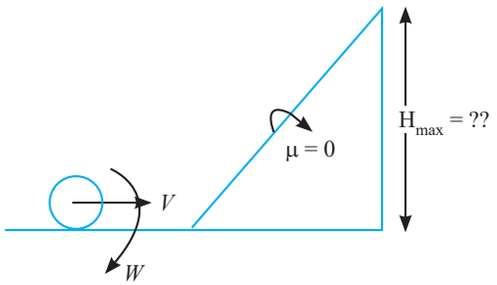
136. On smooth inclined plane, pure rolling motion can't start

(*a*) True (*b*) False

137. Find the time taken by the ball to reach the ground



138. Find maximum height if object is in pure rolling motion.



139. A solid cylinder of mass M and radius R rolls down an inclined plane of height h without slipping. The speed of its centre of mass when it reaches the bottom is

(*a*) $\sqrt{2gh}$ (*b*) $\sqrt{\frac{4}{3}gh}$ (*c*) $\sqrt{\frac{3}{4}gh}$ (*d*) $\sqrt{\frac{4g}{h}}$

140. A solid sphere is rolling without slipping on a level surface at a constant speed of 2.0 ms-1. How far can it roll up a 30° ramp before it stops?

141. A solid sphere is rolling down an inclined plane without slipping of height 20 m. Calculate the maximum velocity with which it will reach the bottom of the plane. ($g = 10\text{m}/\text{s}^2$)

142. Two disc having masses in the ratio 1 : 2 and radii in the ratio 1 : 8 roll down without slipping one by one from an inclined plane of height h. The ratio of their linear velocities on reaching the ground is

(a) 1 : 16 (b) 1 : 128 (c) $1:8\sqrt{2}$ (d) 1 : 1

143. A solid cylinder of mass M and radius R rolls down an inclined plane of height h. The angular velocity of the cylinder when it reaches the bottom of the plane will be

(a) $\frac{1}{2R}\sqrt{gh}$ (b) $\frac{2}{R}\sqrt{gh}$ (c) $\frac{2}{R}\sqrt{\frac{gh}{3}}$ (d) $\frac{2}{R}\sqrt{\frac{gh}{2}}$

144. The speed of a uniform solid cylinder after rolling down an inclined plane of vertical height H, from rest without sliding is

(a) $\sqrt{\frac{gH}{3}}$ (b) $\sqrt{\frac{2gH}{3}}$ (c) $\sqrt{gh}$ (d) $\sqrt{\frac{4gH}{3}}$

145. A thin uniform circular ring is rolling down an inclined plane of inclination 30° without slipping. Its linear acceleration along the inclined plane is

(a) g (b) $\frac{g}{2}$ (c) $\frac{g}{3}$ (d) $\frac{g}{4}$

146. An inclined plane makes an angle of 30° with the horizontal. A solid sphere rolling down this inclined plane from rest without slipping has a linear acceleration equal to

(a) $\frac{g}{3}$ (b) $\frac{2g}{3}$ (c) $\frac{5g}{7}$ (d) $\frac{5g}{14}$

147. The ratio of the accelerations for a solid sphere (mass m and radius R) rolling down an incline of angle 0 without slipping and slipping down the incline without rolling is

148. A hollow cylinder and a solid cylinder are rolling without slipping down an inclined plane, then which of these reaches earlier?

(a) Solid cylinder (b) Hollow cylinder
(c) Both simultaneously (d) Can't say anything

149. A solid sphere and a solid cylinder having the same mass and radius, roll down the same incline. The ratio of their acceleration will be

(a) 15 : 14 (b) 14 : 15 (c) 5 : 3 (d) 3 : 5

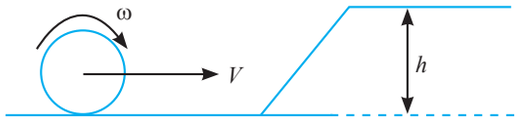
150. A small object of uniform density rolls up a curved surface with an initial velocity v. It reaches upto a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is:

(a) Solid sphere (b) Hollow sphere
(c) Disc (d) Ring

151. Two disc having masses in the ratio 1 : 2 and radii in the ratio 1 : 8 roll down without slipping one by one from an inclined plane of height h. The ratio of their linear velocities on reaching the ground is

(a) 1 : 16 (b) 1 : 128 (c) $1:8\sqrt{2}$ (d) 1 : 1

152. A solid sphere is rolling on a frictionless surface, shown in figure with a translational velocity v ms^{-1}. If it is to climb the inclined surface, then v should be

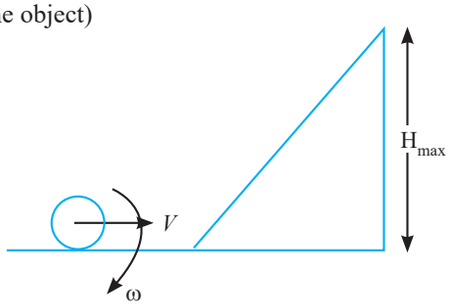


(a) $\geq\sqrt{\frac{10}{7}gh}$ (b) $\geq\sqrt{2gh}$

(c) $\geq\sqrt{\frac{5}{7}gh}$ (d) $\frac{10}{7}gh$

Rolling Motion on Rough Inclined Plane

153. Friction is sufficient then find maximum height attain by the object. (k = radius of gyration of the object z, R = radius of the object)



154. A solid sphere is thrown up a rough incline. The sphere rolls up without slipping and eventually comes down rolling without slipping. The direction of rolling friction in upward and downward motion respectively is

(a) Downward, upward (b) Upward, downward
(c) Downward, downward (d) Upward, upward

155. Two solid spheres of different mass, radii and density roll down a rough inclined plane under identical situation. Their time to come down is independent of their

(a) Mass (b) Radius
(c) Density (d) All of these

156. A thin circular ring first slips down a smooth incline then rolls down a rough incline of identical geometry from same height. Ratio of time taken in the two motion is

(a) $\frac{1}{2}$ (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{4}$

157. When a body is rolling without slipping on a rough horizontal surface, the work done by friction is

(a) Always zero (b) May be zero
(c) Always positive (d) Always negative

158. What is the minimum coefficient of friction for a solid sphere to roll without slipping on an inclined plane of inclination θ?

(a) $\frac{2}{7}\tan\theta$ (b) $\frac{1}{3}g\tan\theta$ (c) $\frac{1}{2}\tan\theta$ (d) $\frac{3u^2}{4g}$

159. A drum of radius R and mass M, rolls down without slipping along an inclined plane of angle θ. The frictional force

(*a*) Dissipates energy as heat

(*b*) Decreases the rotational motion

(*c*) Decreases the rotational and translational motion

(*d*) Converts translational energy to rotational energy

160. A cylinder of radius R and M rolls without slipping down a plane inclined at an angle θ. Coefficient of friction between the cylinder and the plane is μ. For what maximum-inclination 0 will the cylinder roll without slipping?

(*a*) $\tan^{-1}\mu$ (*b*) $\tan^{-1} 2\mu$

(*c*) $\tan^{-1} 3\mu$ (*d*) tan–1 μ/3

161. A solid sphere is rolling down an inclined plane without slipping of height 20 m. Calculate the maximum velocity with which it will reach the bottom of the plane. (g = 10m/s^2)

162. The ratio of the accelerations for a solid sphere (mass m and radius R) rolling down an incline of angle θ without slipping and slipping down the incline without rolling is

163. A disc of mass m and radius r rolls on a horizontal plane without slipping with the speed u. If it starts climbing up an inclined plane, the maximum height it would attain will be

(*a*) $\frac{3u^2}{4g}$ (*b*) $\frac{4u^2}{3g}$ (*c*) $\frac{2u^2}{3g}$ (*d*) $\frac{u^2}{3g}$

164. An initial momentum is imparted to a homogeneous sphere, as a result of which it begins to roll without slipping up an inclined plane at a speed of V_0 = 10 m/s. The plane makes an angle θ = 45° with horizontal. What height h will the sphere rise to? (g = 10 m/S^2)

(*a*) 5 m (*b*) 7 m (*c*) 8 m (*d*) 6 m

165. A solid cylinder is rolling down on an inclined plane of angle θ. The coefficient of static friction between the plane and cylinder is μ_s, then condition for the cylinder not to slip is

(*a*) $\tan\theta \geq \mu_s$ (*b*) $\tan\theta > 4\mu_s$

(*c*) $\tan\theta \leq 3\mu_s$ (*d*) $\tan\theta < \mu_s$

166. Consider a situation in which a ring, a solid cylinder and solid sphere roll down on same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter.

The correct statement for this situation is:

(*a*) All of them will have same velocity.

(*b*) The ring has greatest and cylinder has least velocity of centre of mass at bottom of inclined plane.

(*c*) Sphere has greatest and ring has least velocity of/ com f/COM at bottom of inclined plane.

(*d*) Cylinder has greatest and sphere has least velocity of COM at bottom of inclined plane.

167. A rigid body of mass M and radius R rolls without slipping on an inclined plane of inclination 0, under gravity. Match the type of body column-I with magnitude of the force of friction column-II.

Column-I		Column-II	
A.	For ring	I.	$\frac{Mg\sin\theta}{2.5}$
B.	For solid sphere	II.	$\frac{Mg\sin\theta}{3}$
C.	for solid cylinder	III.	$\frac{Mg\sin\theta}{3.5}$
D.	For hallow cylinder	IV.	$\frac{Mg\sin\theta}{2}$

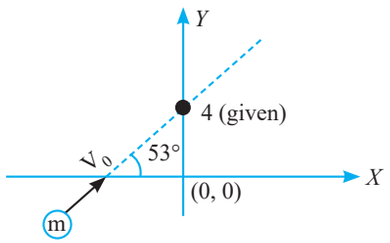
Choose the correct answer from the options given below:

(*a*) A-IV, B-III, C-II, D-IV (*b*) A-I, B-II, C-III, D-IV

(*c*) A-II, B-I, C-IV, D-III (*d*) A-II, B-IV, C-I, D-III

Angular Momentum

168. An object of mass m is moving with velocity v along the path shown in the figure. The object is moving with a velocity vector inclined at an angle of 53° to the horizontal. Calculate the angular momentum of the object about the origin.



169. The position of a panicle is given by $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k})$ and momentum $\vec{P} = (3\hat{i} + 4\hat{j} - 2\hat{k})$. The angular momentum is perpendicular to

(*a*) X-axis

(*b*) Y-axis

(*c*) Z-axis

(*d*) Line at equal angles to all the three axes

170. A force $\vec{F} = a\hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$

The value of a for which angular momentum about origin is conserved is

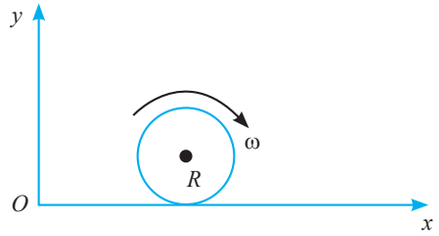
(*a*) 0 (*b*) 1 (*c*) –1 (*d*) 2

171. A particle of mass m is rotating in a plane in circular path of radius r. Its angular momentum is L. The centripetal force acting on the particle is

(*a*) L^2/mr (*b*) L^2m/r^2 (*c*) L^2/m^2r^2 (*d*) L^2/mr^3

172. When a mass is rotating in a plane about a fixed point, its angular momentum is directed along
(*a*) The radius
(*b*) The tangent to orbit
(*c*) Line at an angle of 45° to the plane of rotation
(*d*) The axis of rotation

173. A spherical shell of 1 kg mass and radius R is rolling with angular speedo on horizontal plane. The magnitude of angular moment of the shell about origin O is $\frac{x}{3}R^2\omega$. . The value of x will be :



(*a*) 2 (*b*) 3 (*c*) 5 (*d*) 4

174. Object is projected with speed u at an angle θ then find Angular Momentum after time '*t*'. When it is at maximum height about origin.

175. Object is projected with speed u at an angle θ then find Angular Momentum after time '*t*', when it is about to collide

176. Ball is projected with u at angle θ then find angular momentum at time '*t*' about point of projection

177. Object is moving with velocity $40\sqrt{2}$ m/s on straight line of equation $y = x + 4$ then find angular momentum about origint

178. If $I_1 > I_2$ and kinetic energy is same for both then option for angular momentum
(*a*) $L_1 > L_2$ (*b*) $L_1 < L_2$ (*c*) $L_1 = L_2$

179. Angular moment of a single particle moving with constant speed along circular path:
(*a*) Changes is magnitude but remains same in the direction.
(*b*) Remains same in magnitude and direction.
(*c*) Remains same in magnitude but changes in direction.
(*d*) It is zero.

Conservation of Angular Momentum

180. A swimmer while jumping into river from a height easily forms a loop in air if
(*a*) He pulls his arms and legs in
(*b*) He spreads his arms and legs
(*c*) He keeps himself straight
(*d*) None of these

181. A horizontal disc rotating freely about a vertical axis through its centre makes 90 revolutions per minute. A small piece of wax of mass m falls vertically on the disc and sticks to it at a distance r from the axis. If the number of revolutions per minute reduce to 60, then the moment of inertia of the disc is

(*a*) mr^2 (*b*) $\frac{3}{2}mr^2$ (*c*) $2\,mr^2$ (*d*) $3\,mr^2$

182. A thin circular ring of mass M and radius r is rotating about its axis with constant angular velocity ω. Two object each of mass m are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with angular velocity given by

(*a*) $\frac{(M+2m)\omega}{2m}$ (*b*) $\frac{2M\omega}{M+2m}$

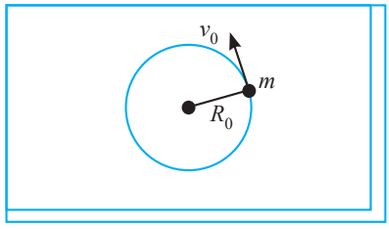
(*c*) $\frac{(M+2m)\omega}{M}$ (*d*) $\frac{M\omega}{M+2m}$

183. If two discs of moment of inertia l_1 and l_2 rotating about colinear axis passing through their centres of mass and perpendicular to their plane with angular speeds ω_1 and ω_2 respectively in opposite directions are made to rotate combined along same axis, then the magnitude of angular velocity of the system is

(*a*) $\frac{l_1\omega_1 + l_2\omega_2}{l_1 + l_2}$ (*b*) $\frac{l_1\omega_1 - l_2\omega_2}{l_1 + l_2}$

(*c*) $\frac{l_1\omega_1 + l_2\omega_2}{\omega_1 + \omega_2}$ (*d*) $\frac{l_1\omega_1 - l_2\omega_2}{\omega_1 - \omega_2}$

184. A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0. The mass is attached to a string which passes through a smooth hole in the plane as shown.
The tension in the string is increased gradually and finally m moves in a circle of radius $R_0/2$. The final value of the kinetic energy is



(*a*) $\frac{1}{2}mv_0^2$ (*b*) mv_0^2 (*c*) $\frac{1}{4}mv_0^2$ (*d*) $2mv_0^2$

185. A heavy solid sphere is thrown on a horizontal rough surface with initial velocity u without rolling. What will be its speed, when it starts pure rolling motion?

(*a*) $\frac{3u}{5}$ (*b*) $\frac{2u}{5}$ (*c*) $\frac{5u}{7}$ (*d*) $\frac{2u}{7}$

186. A uniform sphere is released on a smooth inclined plane from the top. When it moves down its angular momentum is:
(*a*) Conserved about every point
(*b*) Conserved about the point of contact only
(*c*) Conserved about the centre of the sphere only
(*d*) Conserved about any point on a fixed line parallel to the inclined plane and passing through the centre of the ball.

187. An ant is sitting at the edge of a rotating disc. If the ant reaches the other end, after moving along the diameter, the angular velocity of the disc will:-

(*a*) Remain constant

(*b*) First decreases and then increase

(*c*) First increases, then decrease

(*d*) Increase continuously

188. A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere?

(*a*) Rotational kinetic energy(*b*) Angular velocity

(*c*) Angular momentum (*d*) Moment of inertia

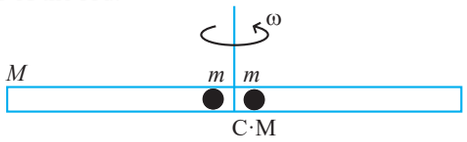
189. A circular platform is mounted on a frictionless vertical axle. Its radius $R = 2$m and its moment inertia about the axle is 200 kgm^2. It is initially at rest. A 50 kg man stands on the edge of the platform and begins to walk along the edge at the speed of 1ms^{-1} relative to the ground. Time taken by the man to complete one revolution is-

(*a*) $\frac{3\pi}{2}s$ (*b*) $2\pi s$ (*c*) $\frac{\pi}{2}s$ (*d*) πs

190. A round disc of moment of inertia I_1 about its axis perpendicular to its plane and passing through its centre is placed over another disc of moment of inertia I_2 rotating with an angular velocity ω_0 about the same axis. The final angular velocity of the combination of discs is

(*a*) $\frac{I_2\omega}{I_1+I_2}$ (*b*) ω (*c*) $\frac{I_1\omega}{I_1+I_2}$ (*d*) $\frac{(I_1+I_2)\omega}{I_1}$

191. A hollow rod of mass M and length L is rotating with an initial angular velocity ω_0 about its center. Two small point masses, each of mass m, are initially positioned at the center of the rod. These point masses begin to move symmetrically away from the center toward the ends of the rod. Find the angular speed ω_f of the system when both masses reach the ends of the rod.



192. A spinning ice skater has an initial kinetic energy $\frac{1}{2}I\omega^2$ She pulls in her outstretched arms, decreasing her moment of inertia by 75%. What is her new angular speed?

(*a*) $\omega/4$ (*b*) $\omega/3$ (*c*) $4\omega/3$ (*d*) 4ω

193. A sphere is released on a smooth inclined plane. Its angular momentum is

(A) Conserved about every point

(B) Conserved about point of contact only

(C) Conserved about centre of sphere only

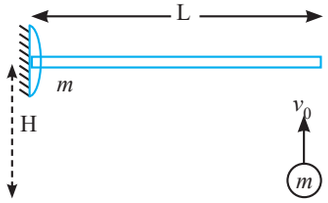
(D) Conserved about any point on a line parallel to inclined plane and passing through centre

From above, true statement/s is/are

(*a*) A, B, C (*b*) Only D (*c*) C and D (*d*) Only C

Collision

194. A uniform rod of mass m in and length L can rotate freely on a smooth horizontal plane about a vertical axis hinged at point H. A point mass having same mass m coming with an initial speed v_0 perpendicular to the rod, strikes the rod in elastically at its free end. Find out the angular velocity of the rod just after collision?.



Loss in K.E.

195. A circular disk of moment of inertia I_t is rotating in a horizontal plane, about its symmetry axis, with a constant angular speed empty set ω_I. Another disk of moment of inertia I_b is dropped coaxially onto the rotating disk. Initially the second disk has zero angular speed. Eventually both the disks rotate with a constant angular speed ω_f . The energy lost by the initially rotating disc to friction is:

(*a*) $\frac{1}{2}\frac{I_b^2}{(I_I+I_b)}\omega_1^2$ (*b*) $\frac{1}{2}\frac{I_t^2}{(I_I+I_b)}\omega_1^2$

(*c*) $\frac{I_b-I_t}{(I_t+I_b)}\omega_1^2$ (*d*) $\frac{1}{2}\frac{I_b\cdot I_t}{(I_t+I_b)}\omega_1^2$

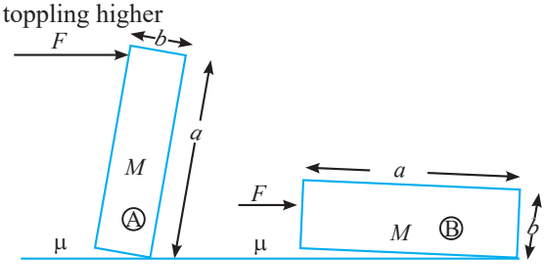
196. Two disc of same M.O.I. is rotating with ω_1 and ω_2 about axis perpendicular to plane. They brought into contact force to face. Then loss in K.E. will be

(*a*) $\frac{1}{2}I(\omega_1+\omega_2)^2$ (*b*) $\frac{1}{4}I(\omega_1-\omega_2)^2$

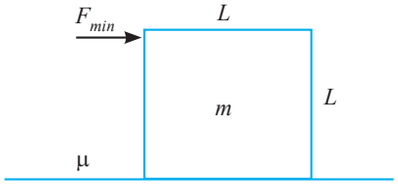
(*c*) $I(\omega_1+\omega_2)^2$ (*d*) $\frac{1}{8}I(\omega_1-\omega_2)^2$

Toppling

197. Two objects, A and B, each of mass m, are subjected to a horizontal force F as shown in the figure. Object A is in an upright position, while object B is lying flat on a frictional surface. The center of mass of both objects is located at height $a/2$ from the base. In which case is the probability of toppling higher

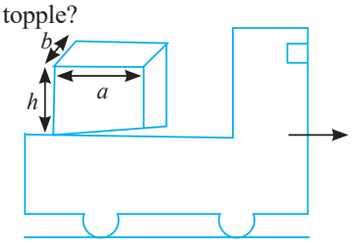


198. A cubical block of mass m and side L rests on a rough horizontal surface with coefficient of friction μ. A horizontal face F is applied on the block as shown. If the coefficient of friction is sufficiently high so that the block does not slide before toppling, the minimum force required to topple the block is:



(*a*) Infinitesimal (*b*) $mg/2$ (*c*) $mg/4$ (*d*) $mg(1-\mu)$

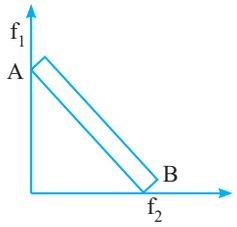
199. What will be the value of maximum acceleration of the truck in the forward direction so that the block kept on the back does not topple?



(*a*) ag/h (*b*) hg/a (*c*) $ag/2h$ (*d*) h/bg

200. A uniform cube of side '*a*' and mass m rests on a rough horizontal table. A horizontal force F is applied normal to one of the faces at a point directly $\frac{3a}{4}$ above the center of the face, at a height above the base. What is the minimum value of F for which the cube begins to tip about on edge ?

201. Wall and floor is rough. Rod is about to slide. Then find correct statement from the following (Given f_1 and f_2 are friction forces)



(A) f_1 is in upward direction and f_2 towards the wall

(B) Normal on B is less then mg

(C) Normal on B is greater then mg

(D) Normal on B is equal to mg

(*a*) A and B are true

(*b*) A and C are true

(*c*) A and D are true

(*d*) Only D is correct

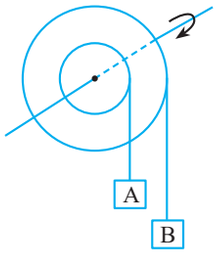
MR* CORNER

1. **Statement I:** Let $\vec{A}$ be a unit vector along axis of Rotation of purely rotating Body and $\vec{B}$ be the unit vector along the resultant force on particle p of Body away from axis. The Value of $\vec{A}.\vec{B}$ will be Zero.

Statement II: If a particle moves with the Constant velocity parallel to the x-axis, then Its angular momentum with respect to the origin will continuously goes on increasing.

2. **Statement I:** A Body is in pure rotation. The linear speed v of a particle, the distance r of the particle from axis, angular velocity ω of the body are related as $\omega = \frac{v}{r}$. Therefore a particle which is located far away from axis will rotate with slower Angular velocity as compared to particle located near the axis of Rotation.

Statement II: Given figure shows a small wheel fixed coaxially on Bigger one of double the radius. The system rotates about common axis. The strings Supporting A and B do not slip on the wheels. If x and y be the distances travelled by A ond B in same time interval then $y = 2x$.



3. **Statement I:** If a Body is rotating uniformly about a vertical axis fixed. in an inertial frame, Then resultant force on a particle of Body which is not on the axis will be Horizontal and intersecting the axis.

Statement II: If a Body is rotating non-uniformly about a vertical axis fixed in an inertial frame, Then the Resultant force on particle of Body which is not present on the axis will be Horizontal and will skew (neither parallel nor intersecting) with the axis.

4. **Statement I:** One end of uniform rod of mass m & length l is clamped. The rod lies on smooth horizontal surface and rotates on it about the clamped end at uniform angular velocity ω The force exerted by the clamp on rod will have horizontal component of $m\omega^2 l$.

Statement II: A circular disc A of radius r is made from an iron plate of Thickness t and another circular disc B of radius 4r is made from an iron plate of Thickness t/4. Equal Torques act on disc A and B, initially both being at rest. At a later instant the linear speeds of point on the rim of A and another point on grim of B will have relation of $V_A > V_B$.

5. **Statement I:** A closed cylindrical tube containing some water not filling entire tube) lies in Horizontal plane. If Tube is rotated about a perpendicular bisector, the moment of inertia of water about the axis will increase.

   Statement II: Let I_1 and I_2 be the moments of inertia of Two bodies of identical geometrical shape. The first made of aluminium and second of Iron I_1 will be equal To I_2.

6. **Statement I:** A cubical block of mass M and edge a slides down a rough inclined plane of inclination θ with a uniform velocity. The Torque of the Normal force on the block about its centre will be zero as Normal will pass through centre of Block.

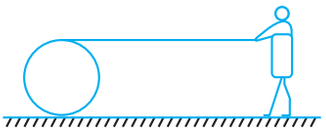
   Statement II: If A person sitting firmly over a rotating stool has his arms Stretched. If he folds his arms, his angular momentam about axis of rotation will increase as Moment of inertia about axis decreases.

7. The Centre of a wheel rolling on a plane surface moves with a speed $V_\circ$. A particle on the rim of wheel at some level as the centre will be moving at speed___.

8. **Statement I:** A solid sphere, a hollow sphere and a disc, all having same mass and Radius, are placed at the top of a smooth incline and released. least Time will be Taken in reaching the bottom is by the solid sphere.

   Statement II: The axis of Rotation of purely rotating body Must Pass through the Centre of mass.

9. A string of negligible Thickness is wrapped several times around a cylinder kept on rough horizontal surface. A man standing at a distance _____ from the cylinder holds one end of the string and pulls cylinder towards him. There is no slipping anywhere The length of string passed through the hand of man while the cylinder reaches his hands is 10 metre.



10. **Statement I:** A particle moves on a Straight line with a uniform velocity. Its angular momentiem is zero about a point on the straight line.

    Statement II: A particle moves. on a straight line with a uniform velocity. Its angular momentum about any given point will remain constant.

11. **Statement I:** If there is no external force acting on a nonrigid body, Angular momentum must Remain constant.

    Statement II: If there is no external force acting on a nonrigid body, Its moment of inertia must Remain constant.

12. **Statement I:** I_A and I_B are the moment of inertia of body about two axes A and B Respectively. The axis A passes through centre of mass of Body but B does not. If I_A is lesser than I_B then axis must be parallel to each other.

    Statement II: I_A and I_B are the moment of inertia of body about two axes A and B Respectively. The axis A passes through centre of mass of Body but B does not. If the axes are parallel to each other. then I_A will be lesser than I_B.

13. **Statement I:** If A sphere is rotating about a diameter, All the particles on the surface will have some linear speed.

    Statement II: Two uniform solid spheres having unequal masses and unequal radii are released from rest from the same height on a rough incline. If the spheres roll without slipping, the two spheres will reach the bottom Together.

14. **Statement I:** A sphere cannot roll on a smooth horizontal surface.

    Statement II: A sphere can roll on surface inclined at an angle θ if the friction coefficient is more than 2/7 tan θ Suppose the friction coefficient is 1/7 tan θ then sphere will make pure Translational motion if a sphere is released from rest on the incline.

Answer Key

1. (a)	2. $I_1 > I_2$	3. $r_\perp = 2$ m, $I = 8$ kg. m^2	4. (d)	5. $\left(\frac{m_1 m_2}{m_1 + m_2}\right) r^2$	6. (d)	7. (c)	8. (5)		
9. (d)	10. (c)	11. (d)	12. (d)	13. (c)	14. (a)	15. (c)	16. (b)	17. (a)	18. (d)
19. (c)	20. (b)	21. (b)	22. (b)	23. (b)	24. (a)	25. (a)	26. (b)	27. (b)	28. (d)
29. $\frac{3}{2}MR^2$	30. $\frac{M_0 L^2}{8}$	31. (a)	32. (a)	33. (c)	34. (c)	35. (a)	36. (c)	37. (c)	38. (c)
39. (a)	40. (a)	41. (9)	42. (a)	43. (b)	44. (a)	45. (b)	46. (c)	47. (d)	48. (a)
49. (b)	50. (5)	51. (a)	52. (d)	53. (c)	54. (b)	55. (d)	56. (c)	57. (d)	58. (b)

59. (*i*) $mu^2 \sin\theta \cos\theta$, (*ii*) $mu^2 \sin(2\theta)$, (*iii*) $mgu\cos\theta t$ **60.** (*c*) **61.** (*b*) **62.** $\frac{mgL\sin\theta}{2}$ **63.** $\tau = FR\otimes$ **64.** (*d*)

65. (*b*) **66.** (*a*) **67.** (*b*) **68.** (*a*) **69.** (*b*) **70.** (*c*) **71.** (*c*) **72.** (*b*) **73.** (*d*)

74. $N_A = \frac{22}{5}kN,\ N_B = \frac{38}{5}kN$ **75.** $\frac{1}{12}$kg **76.** (*a*) **77.** (*a*) **78.** 1.5 rad/s^2 **79.** 100 sec **80.** 37.5

81. (*d*) **82.** (*a*) **83.** (*a*) **84.** (*a*) **85.** (1 rad/s) **86.** (*b*) **87.** (*a*) **88.** 0.1 sec **89.** (*a*) **90.** 728

91. (*c*) **92.** (*b*) **93.** (*a*) **94.** (*d*) **95.** (*c*) **96.** (*a*) **97.** (*c*) **98.** (*d*) **99.** (*b*) **100.** (*c*)

101. $\vec{a}_{CM} = \frac{3}{4}g,\ \vec{a}_P = 0,\ \vec{a}_Q = \frac{3g}{2},\ N_p = \frac{mg}{4}$ **102.** (*a*) **103.** 100 rad/s **104.** (*b*)

105. $a = \frac{(m_2 - m_1)g}{\left(m_1 + m_2 + \frac{I}{R^2}\right)},\ \alpha = \frac{a}{R} = \frac{(m_2 - m_1)g}{\left(m_1 + m_2 + \frac{I}{R^2}\right)R}$ **106.** (*a*) **107.** (5 : 3) **108.** $v = \sqrt{3gl}$

109. (*b*) **110.** (*b*) **111.** (*a*) **112.** (*a*) **113.** Hollow sphere **114.** $\sqrt{\frac{20}{3}}\ m/s$ **115.** (*c*)

116. (*b*) **117.** (*a*) **118.** (*a*) **119.** (*a*) **120.** (*a*) **121.** (*d*) **122.** 6J **123.** K.E = 3125 joule **124.** (*c*)

125. (*d*) **126.** (*c*) **127.** (*b*) **128.** (*d*) **129.** (*a*) **130.** (*d*) **131.** (*a*) **132.** $\frac{2F}{5}$ **133.** (9) **134.** (*a*)

135. (*b*) **136.** (*a*) **137.** $T = \sqrt{\frac{2H}{g}}$ **138.** $H = \frac{V^2}{2g}$ **139.** (*b*) **140.** 56 cm **141.** $V = 20\sqrt{\frac{5}{7}}$m/s

142. (*d*) **143.** (*c*) **144.** (*d*) **145.** (*d*) **146.** (*d*) **147.** 5/7 **148.** (*a*) **149.** (*a*) **150.** (*c*) **151.** (*d*)

152. (*a*) **153.** $H = \frac{V^2}{2g}\left(\frac{R^2 + k^2}{R^2}\right)$ **154.** (*d*) **155.** (*d*) **156.** (*c*) **157.** (*a*) **158.** (*a*) **159.** (*b*) **160.** (*c*)

161. $\sqrt{\frac{2000}{7}}$ **162.** $\frac{5}{7}$ **163.** (*a*) **164.** (*b*) **165.** (*c*) **166.** (*c*) **167.** (*a*) **168.** $L = \frac{12mV}{5}$ **169.** (*a*)

170. (*c*) **171.** (*d*) **172.** (*d*) **173.** (*c*) **174.** $L = \frac{mu^3 \sin^2\theta\cos\theta}{2g}$ **175.** $L = \frac{2mu^3 \sin^2\theta\cos\theta}{g}$ **176.** $L = \frac{mgu\cos\theta t^2}{2}$

177. 160 kg. *m/s* **178.** (*a*) **179.** (*b*) **180.** (*a*) **181.** (*b*) **182.** (*d*) **183.** (*b*) **184.** (*d*) **185.** (*c*)

186. (*d*) **187.** (*c*) **188.** (*c*) **189.** (*b*) **190.** (*a*) **191.** $\omega_O = \frac{M\omega}{M + 6m}$ **192.** (*d*) **193.** (*b*) **194.** $3v_0/4L$

195. (*d*) **196.** (*b*) **197.** For object A **198.** (*b*) **199.** (*a*) **200.** $\frac{2mg}{3}$ **201.** (*a*)

MR* CORNER

1. I. True, II. False **2.** I. False, II. True **3.** Both are True **4.** I. False, II. True **5.** I. True, II. False
6. Both False **7.** $\sqrt{2}$ V° **8.** Both are false **9.** 5 metre. **10.** Both are True
11. I. True, II. False **12.** I. False, II. True **13.** I. False, II. True **14.** I. Both are false